LIGHTYEAR

MS92 Frozen factory repeatability experiment

Word edition

Status: Completed experiment; zero verified successes in 40 planned trials. **Date:** 28 September 2026.

MS92 tested whether the previous single successful journey would repeat under frozen conditions. All 40 slots ran and halted. This establishes that the tested factory configuration was not reliable enough to claim repeatable autonomous delivery. It does not establish that reliable delivery is impossible: the experiment also exposed defects in the judge, public contract and execution equipment that MS93 addresses separately.

Experiment and observed outcomes

Ten fresh generations ran in each of four conditions. Baseline repeated the operations journey; procure-to-pay introduced purchasing; the other two conditions removed the decimal-format instruction or Boolean hint separately. Conditions were interleaved. Each trial had at most five client calls; the cohort allowed 200 calls and 40 local execution hours. Its declaration, controller, judge and inputs were fixed before execution. No trial was replaced or restarted.

Condition	Observed result
Baseline	0 first-try passes; 0 repaired passes; 10 halts
Procure to pay	0 first-try passes; 0 repaired passes; 10 halts
Without decimal instruction	0 first-try passes; 0 repaired passes; 10 halts
Without Boolean hint	0 first-try passes; 0 repaired passes; 10 halts

Every condition had zero invalid slots and zero unstarted slots under the original controller's classification. Across the cohort, 38 trials halted without supported repair feedback and two exhausted their frozen call limit. Twenty-six trials used repairs, producing 38 revised candidates. None passed the combined gate.

The observed pass rate was 0 of 10 in each condition. The report's descriptive 95 percent Wilson interval is 0 to about 27.8 percent per condition under a binomial working model. Small samples, shared inputs and provider drift limit generalization. Conditions must not be pooled to claim a general factory success rate or to estimate the causal value of either removed hint.

Cost and interpretation

Measurement	Recorded value
Client invocations	116 total: 78 builder and 38 analyst
Native attempts	78, all unsuccessful under the frozen gate
Input tokens	2,435,584
Output tokens	583,421
Cached input tokens	373,888, included in input tokens
Summed trial elapsed time	37,832.517 seconds, approximately 10.51 hours
Human candidate repair bytes	Zero
Mid-campaign controller changes	Zero
Dollar billing	Unavailable through the signed-in transport
Cloud resources	None; local Docker only

Baseline consumed 22 calls and 16 native attempts; purchasing 36 and 23; without-decimal 32 and 21; without-boolean 26 and 18. Client invocations are the accounting unit; internal provider retries and a

provider-resolved model snapshot are not exposed. The requested model was gpt-6-astra with high reasoning, through the pinned Codex CLI. Local electricity and CPU cost were not metered.

Why the halts matter

Fifty-five native attempts ended in harness exceptions and 23 in gate failures. None produced a complete combined gate file under the old implementation. Signed publications establish retained artifact identity and provenance; a verified archive is not a passed native comparison. The original controller's zero-invalid count does not mean every judge execution completed normally.

Investigation found ambiguous invoice-line assumptions, purchasing costing side effects, missing or duplicate trace entries and compiler feedback the exporter did not recognize. A halted candidate therefore cannot automatically be described as a proven business defect. MS92 preserves the original classifications; MS93 introduces explicit execution-failure, judge-error and insufficient-evidence results for subsequent work.

Builder input provenance

Baseline inherited MS86's API and locking examples, MS88/MS89's Boolean, posting-signature and no-reposting lessons, and MS89/MS90's decimal-format lesson. Purchasing used the predeclared purchasing contract. The ablations removed one named instruction each; no campaign-specific lesson was added after generation began. Zero repair bytes excludes development of those inherited inputs and of the judge and controller.

Evidence and closeout

Frozen cohort plan SHA-256:

`b38d5979937e37c7b2d1072dae56f38923739b8f77449bef7db5f99176157d4b`

Signed rate report SHA-256:

`7486b0839125ad79571221379e796609b1c1d8a417533d021ca9e514cc725973`

All 40 signed publications were verified at termination, and all 78 cleanup receipts verified. The campaign left no owned containers, networks or volumes. Nine unrelated retained volumes were explicitly excluded from cleanup. Monitoring was stopped after the terminal report; the cohort has no resume or amendment path.

The repository contains the cohort report, declaration, trial plans and receipts, publication receipts, terminal verification records and signed archive index. The 40 complete archives total 2,499,047,692 bytes and remain at the recorded local publication paths. They are not embedded in this PR. A fresh clone can inspect and authenticate the compact results; full artifact reconstruction requires those retained archives and the trusted operator key. Signatures are attribution and tamper evidence, not independent attestation.

Commercial significance

The experiment replaces a single success story with a reproducibility measurement and preserves failures that can guide development. It does not support an autonomous production-readiness claim. The next engineering priority is to qualify the judge and execution environment with a known reference and deliberate faults before measuring the factory again.

References

- Signed rate report
- Evidence scope and archive index
- Builder input provenance
- MS93 judge and environment qualification